

Early Detection of Parkinson’s Disease through Machine Learning and Deep Learning Models

Akshay Channam, Addanki Naveen, Ram Swarup Bavikati,
Bhagath Singh Bondili, Sree Venkat Chintakula

Affiliation {achannam, naddank, rbavikat, bbondili, sreechin}@iu.edu

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Abstract

Parkinson’s Disease (PD) is a progressive neurodegenerative disorder characterized by the gradual degeneration of dopamine-producing neurons. It presents with motor symptoms such as tremors, rigidity, bradykinesia, and postural instability, alongside non-motor symptoms, including speech impairments. Speech disturbances, which affect up to 90% of PD patients, often emerge early in the disease course, making them valuable biomarkers for early detection. This research explores automated voice analysis, combining machine learning (ML) and deep learning (DL) approaches to develop a non-invasive, cost-effective, and accurate system for early PD diagnosis. By employing tools such as OpenSMILE, Parselmouth(Praat), and LIBROSA for feature extraction and leveraging three distinct methodologies across two datasets, this study achieves remarkable accuracy in real-time PD classification. The findings underscore the transformative potential of speech analysis in clinical diagnostics.

Keywords

Parkinson’s Disease, Speech Analysis, Machine Learning, Deep Learning, Biomarkers

1 Introduction

1.1 Parkinson’s Disease Overview

Parkinson’s Disease (PD) is a debilitating neurodegenerative disorder that affects millions of people worldwide. It primarily impacts dopamine-producing neurons in the brain, leading to progressive motor symptoms such as tremors, rigidity, and bradykinesia. Beyond motor symptoms, non-motor features like speech impairments often precede clinical diagnosis by several years. Speech-related symptoms include hypophonia (reduced loudness), monotone pitch, slurred articulation, and inappropriate pauses. These early-stage manifestations highlight the potential of speech as a critical biomarker for PD detection Little et al. [2007], Tsanas et al. [2012].

1.2 Problem Statement

Current diagnostic techniques for PD rely on clinical evaluations, imaging technologies, and laboratory tests. These approaches are not only invasive and expensive but also ineffective for early detection. By the time PD is clinically diagnosed, significant neuronal damage has already occurred. Speech disturbances, present in approximately 90% of PD cases, offer a promising avenue for early,

non-invasive diagnosis. Despite this potential, there is a lack of automated and scalable systems for analyzing speech features, hindering clinical adoption.

1.3 Objective

The primary objective of this research is to develop an automated framework for early detection of PD by analyzing speech features. The framework combines feature-based machine learning (ML) and deep learning (DL) models, providing robust classification capabilities and real-time diagnostic potential.

1.4 Scope of Study

This study leverages publicly available datasets and open-source tools to create a scalable framework for PD detection. By integrating diverse methodologies, the research emphasizes accuracy, accessibility, and adaptability, paving the way for real-world implementation.

2 Datasets

2.1 Acoustic Features Dataset

The first dataset includes 81 samples, with 40 recordings from PD patients and 41 from healthy controls. Participants performed sustained vowel phonations, which provided a basis for extracting acoustic features such as jitter, shimmer, and harmonic-to-noise ratio (HNR). These features were analyzed to identify characteristic speech patterns associated with PD.

2.2 Italian Parkinson’s Voice and Speech Dataset

The second dataset comprises 495 recordings from 65 participants. This dataset includes a variety of speech tests, such as vowel pronunciations and complex speaking tasks, enabling a comprehensive evaluation of speech impairments in PD patients from Dimauro and Girardi.

2.3 Preprocessing

Both datasets were subjected to the following preprocessing steps:

- **Normalization:** Standardized features to ensure consistency across samples.
- **Noise Reduction:** Applied filtering techniques to remove background noise and enhance signal quality.
- **Segmentation:** Divided audio samples into smaller units to improve the granularity of feature extraction.

3 Comprehensive Speech Features

Speech analysis involves several feature categories, each capturing unique characteristics of vocal patterns. Below is an in-depth overview of key speech features:

3.1 Acoustic Features

- **Fundamental Frequency (F0):** Represents the pitch of the voice, often reduced in PD patients.
- **Jitter:** Captures frequency variation, indicating vocal instability, which is elevated in PD patients.
- **Shimmer:** Reflects amplitude variation, commonly observed in PD patients as irregular vocal intensity.
- **Harmonic-to-Noise Ratio (HNR):** Measures voice clarity, with lower values indicating degraded voice quality in PD.

3.2 Spectral Features

- **Mel Frequency Cepstral Coefficients (MFCCs):** Capture spectral characteristics, critical for identifying articulation changes.
- **Spectral Slope and Entropy:** Indicate the shape and randomness of the power spectrum, reflecting vocal tone.
- **Zero-Crossing Rate (ZCR):** Measures the rate of signal sign changes, correlating with vocal noisiness.

3.3 Prosodic Features

- **Energy Dynamics:** Analyze loudness variations over time.
- **Speech Tempo:** Captures speaking rate, often slower in PD patients.
- **Rhythm and Intonation:** Represent variations in speech melody and stress, diminished in PD.

3.4 Temporal Features

- **Pause Duration and Frequency:** PD patients often exhibit prolonged pauses due to speech planning difficulties.
- **Segment Duration:** Measures continuous speech length, usually shorter in PD.

3.5 Features Used in This Study

From these categories, 52 key features were selected for their discriminative power in detecting PD. These include:

1. **os_audspec_lengthL1norm_sma_upleveltime50:** Measures the time taken for the audio signal to reach 50% of its maximum level.
2. **total_features:** Total count of extracted features.
3. **praat_count:** Number of features derived from Praat analysis.

4. **os_pcm_fftMag_spectralSlope_sma_upleveltime90:** Measures the spectral slope at 90% energy level.
5. **os_audSpec_Rfilt_sma_de_4_maxPos:** Maximum position of energy in a filtered frequency band.
6. **os_pcm_zcr_sma_min:** Minimum value of zero-crossing rate.
7. **os_audspec_lengthL1norm_sma_kurtosis:** Kurtosis of the spectral length, capturing distribution asymmetry.
8. **lb_zcr_mean:** Average zero-crossing rate, indicating signal noisiness.
9. **os_audspec_lengthL1norm_sma_skewness:** Skewness of the spectral length distribution, reflecting its asymmetry.
10. **os_pcm_fftMag_spectralEntropy_sma_qregc2:** Quantifies spectral randomness using entropy.
11. **os_pcm_zcr_sma_linregc2:** Linear regression coefficient for zero-crossing rate over time.
12. **lb_global_mfcc_mean:** Average MFCC values, crucial for capturing spectral patterns.
13. **os_logHNR_sma_de_lpc1:** Linear predictive coding for Harmonics-to-Noise Ratio, capturing voice quality.
14. **os_audspec_lengthL1norm_sma_upleveltime75:** Time for 75% of spectral energy to be reached.
15. **os_pcm_loudness_sma_de_amean:** Average loudness of the audio signal.
16. **os_audspec_lengthL1norm_sma_maxPos:** Maximum position of spectral energy.
17. **os_pcm_fftMag_spectralVariance_sma_range:** Range of spectral variance across frequency bands.
18. **os_audSpec_Rfilt_sma_24_quartile2:** Second quartile of energy distribution in a filtered band.
19. **os_audspec_lengthL1norm_sma_percentile1.0:** First percentile of spectral length distribution.
20. **os_voiceProb_sma_linregc1:** Probability of voiced regions based on linear regression.
21. **os_audSpec_Rfilt_sma_de_1_maxPos:** Maximum position of energy for the first filtered band.
22. **os_audspec_lengthL1norm_sma_lpgain:** Gain coefficient from spectral length prediction.

These features represent a combination of spectral, prosodic, and temporal properties, capturing critical aspects of speech affected by Parkinson’s disease.

4 Feature Relationships

Figures 1 through 4 depict key feature relationships, showcasing the separation between Healthy Controls (HC) and Parkinson's Disease (PD) patients.

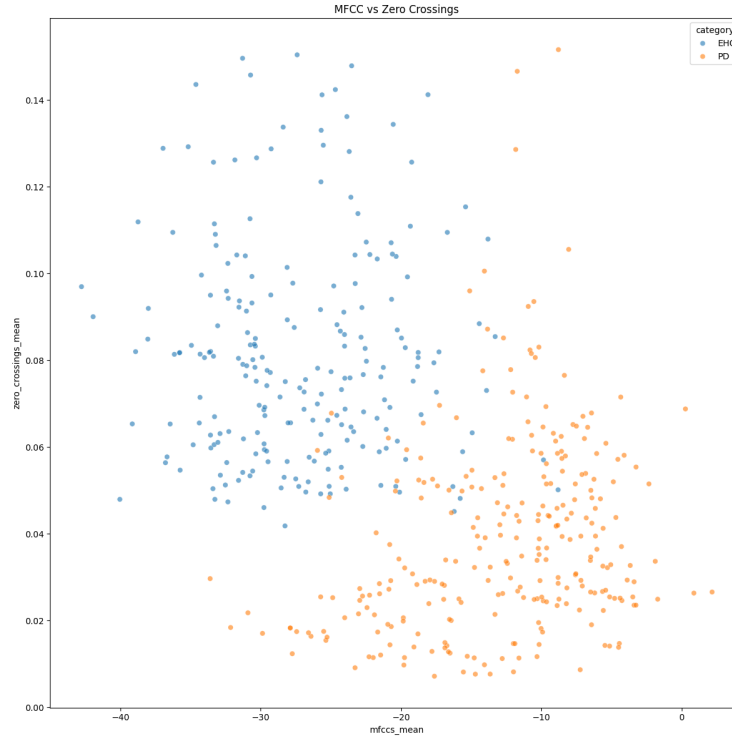


Figure 1: MFCC Mean vs Zero-Crossing Rate: Clear separation between HC and PD. Lower MFCC values and distinct ZCR patterns reflect vocal articulation changes.

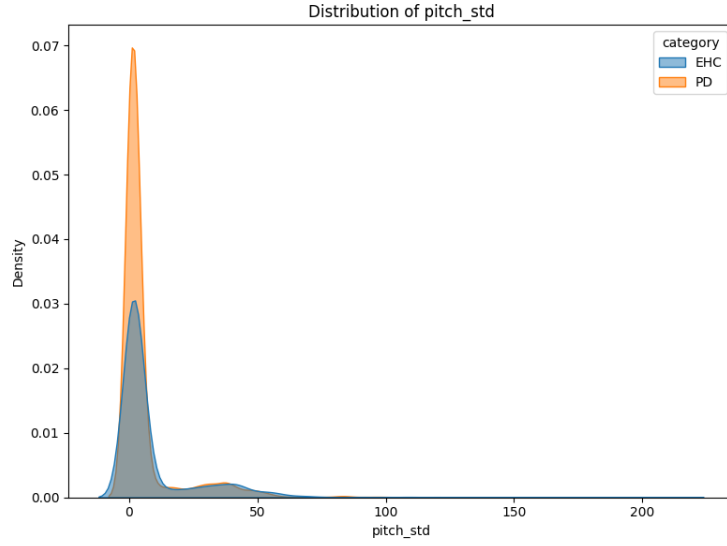


Figure 2: Pitch variability comparison between Healthy Controls and PD patients. PD patients exhibit reduced variability, indicating monotonic speech patterns.

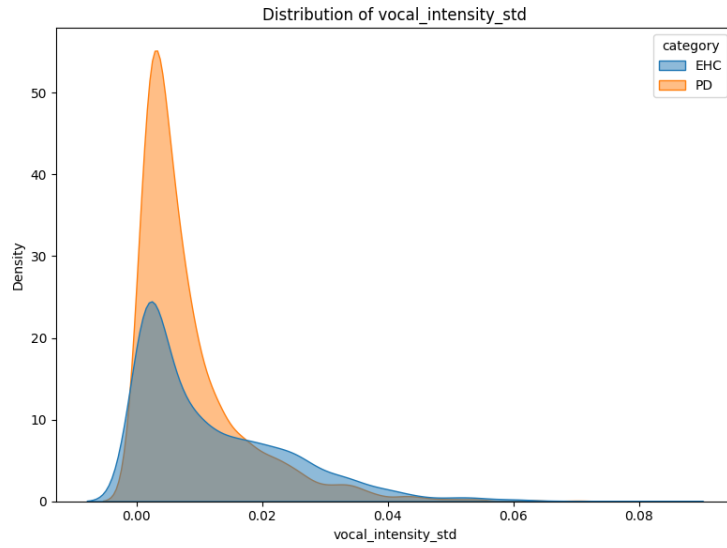


Figure 3: Comparison of vocal intensity levels. PD patients demonstrate reduced intensity due to diminished vocal strength.

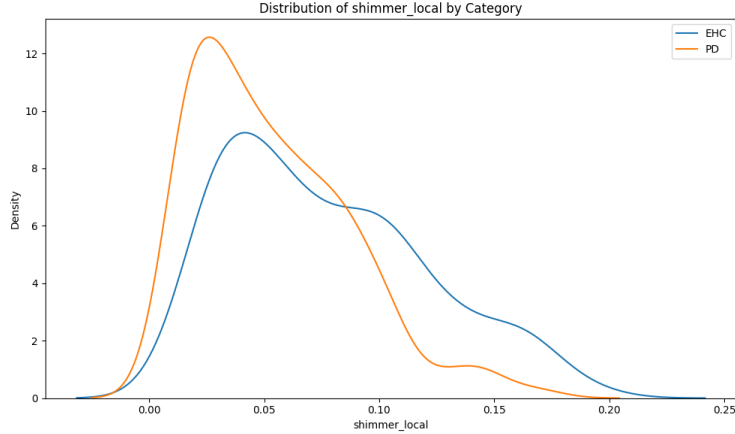


Figure 4: Shimmer levels in Healthy Controls versus PD patients. Elevated shimmer in PD patients indicate vocal instability.

5 Tools and Libraries

5.1 OpenSMILE

OpenSMILE (Open Speech and Music Interpretation by Large-space Extraction) is a robust and widely used toolkit for extracting low-level descriptors (LLDs) and functional features from audio signals OpenSMILE Toolkit. It provides:

5.1.1 Feature Sets

1. **ComParE_2016:** Extracts 6,373 features, including MFCCs, voicing parameters, and spectral characteristics, suitable for detailed paralinguistic analysis.
2. **eGeMAPS:** A compact set of 88 features, emphasizing frequency, energy, and spectral parameters for clinical applications.
3. **GeMAPS:** Offers 62 features optimized for resource-constrained environments, enabling real-time analysis.

5.1.2 Applications

OpenSMILE has been extensively applied in clinical diagnostics, emotion recognition, and real-time speech analysis. In this study, OpenSMILE served as the primary tool for extracting phonatory, spectral, and temporal features, forming the foundation for ML and DL model training.

5.1.3 Integration in Methodologies

OpenSMILE was utilized in all methodologies to extract relevant features from audio data. These features were essential for both traditional ML models and DL models, ensuring consistency in analysis across datasets.

5.2 Librosa

Librosa is a Python package for music and audio analysis Librosa. It provides:

5.2.1 Feature Sets

1. **Time-Domain Features:** Extracts zero-crossing rate (`lb_zcr_mean`) and other temporal characteristics for analyzing signal patterns.
2. **Spectral Features:** Computes Mel-frequency cepstral coefficients (`lb_global_mfcc_mean`) and spectral contrast for detailed frequency analysis.
3. **Statistical Measures:** Calculates various statistical features like minimum values of global MFCCs (`lb_global_mfcc_mean_min`) for comprehensive signal characterization.

5.2.2 Applications

Librosa was instrumental in our study for spectral analysis and feature extraction, particularly in characterizing the frequency content and distribution of voice signals. These features proved crucial in distinguishing between PD and healthy control samples.

5.2.3 Integration in Methodologies

Librosa was used alongside OpenSMILE and Parselmouth to extract complementary features from audio data. These features contributed to both our initial feature set of 7,195 features and our final optimized set of 52 features used in the LightGBM model.

5.3 Parselmouth

Parselmouth is a Python interface to the Praat phonetic analysis software Parselmouth. It provides:

5.3.1 Feature Sets

1. **Voice Quality Features:** Extracts fundamental frequency (F0) and harmonics-to-noise ratio (HNR) for voice quality assessment.
2. **Prosodic Features:** Captures pitch variations and formant-related features for intonation analysis.
3. **Clinical Metrics:** Computes measurements particularly relevant for clinical voice analysis, including jitter and shimmer.

5.3.2 Applications

In our research, Parselmouth served as a specialized tool for extracting precise voice quality measurements, particularly those known to be affected in Parkinson’s disease. Its clinical-grade measurements provided robust features for our machine learning models.

5.3.3 Integration in Methodologies

Parselmouth was integrated into our feature extraction pipeline to provide high-precision voice measurements. These features were combined with those from OpenSMILE and Librosa to create a comprehensive feature set that achieved high accuracy in PD detection across multiple validation datasets.

6 Implementations

6.1 Implementation 1: Feature-Based Machine Learning Models

6.1.1 Model Architecture

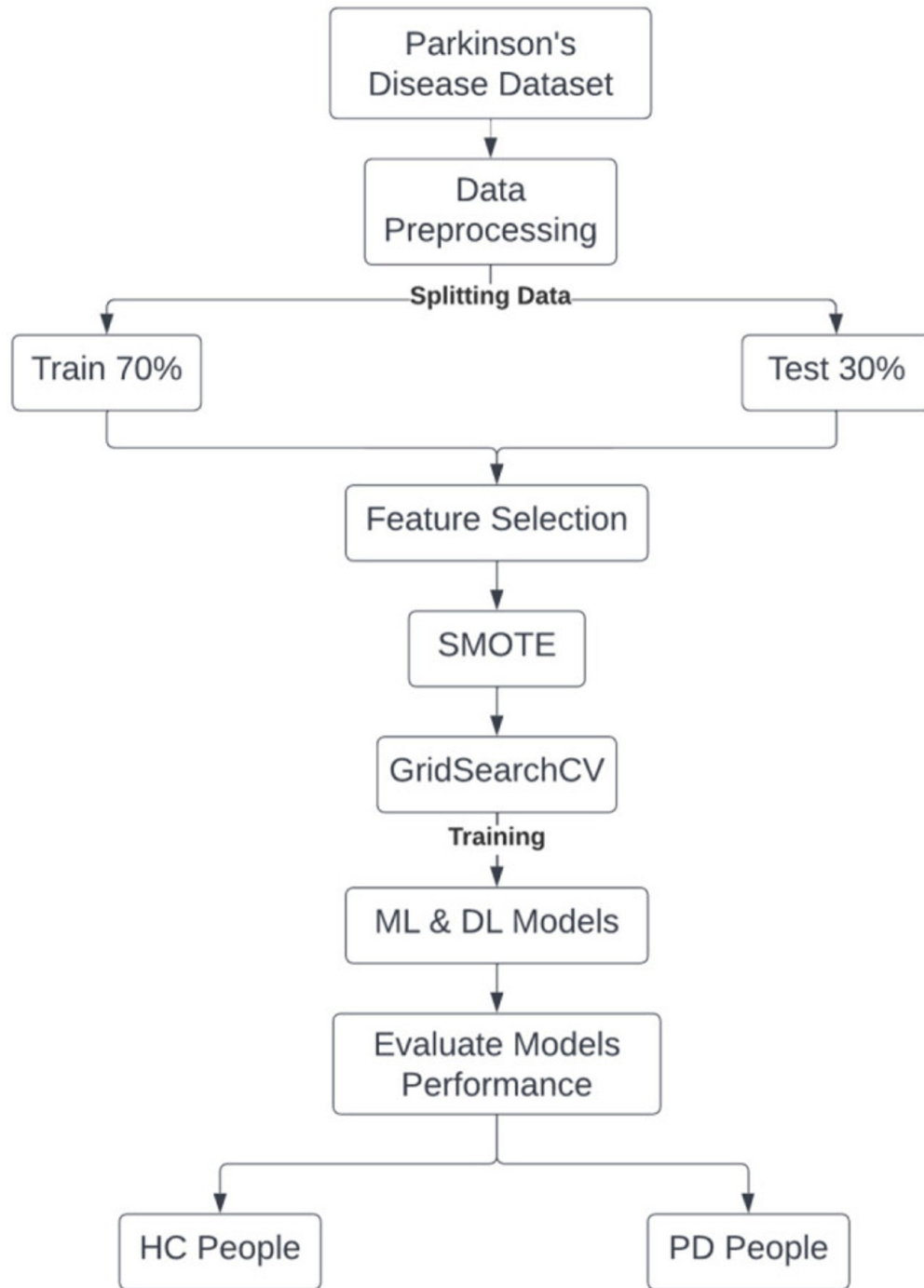


Figure 5: Model Architecture for Implementation 1,3

6.1.2 Feature Extraction

Using OpenSMILE and Praat, key features such as jitter, shimmer, HNR, MFCCs, alpha ratio, and spectral centroid were extracted. These features provided a comprehensive representation of speech impairments associated with PD Alshammri et al. [2023].

6.1.3 Preprocessing

1. **Class Balancing:** The datasets exhibited imbalances between PD and healthy samples. To address this, the Synthetic Minority Oversampling Technique (SMOTE) was applied, ensuring equitable representation across classes.
2. **Standardization:** Features were normalized using StandardScaler to maintain uniform scaling across all data points.

6.1.4 Feature Reduction

1. **SelectKBest:** This statistical technique identified the top features most relevant to classification, including MDVP:F0 (mean fundamental frequency) and shimmer.
2. **Principal Component Analysis (PCA):** PCA was employed to reduce dimensionality while preserving 95% of the data's variance.

6.1.5 Model Development

Several machine learning models were trained and evaluated, including:

- **Support Vector Machine (SVM):** Utilized a radial basis function kernel for non-linear classification.
- **Random Forest (RF):** Built an ensemble of decision trees to enhance robustness and generalizability.
- **K-Nearest Neighbors (KNN):** Classified instances based on proximity in the feature space.
- **Decision Tree (DT):** Offered interpretable rule-based hierarchical classification.
- **Multi-Layer Perceptron (MLP):** Achieved the highest accuracy (84%) by leveraging dense neural connections to learn complex patterns in the data.

6.1.6 Model Training and Hyperparameter Tuning

Hyperparameters for each model were optimized using GridSearchCV. For instance:

- **SVM:** Tuned kernel parameters to improve decision boundary flexibility.
- **RF:** Adjusted tree depth and the number of estimators for better ensemble performance.
- **MLP:** Fine-tuned the number of layers, neurons per layer, and activation functions.

6.1.7 Evaluation Metrics

Each model was evaluated using the following metrics:

- **Accuracy:** Percentage of correct predictions.
- **Precision:** Proportion of true positive predictions among all positive predictions.
- **Recall (Sensitivity):** Proportion of actual positives correctly identified.
- **F1-Score:** Harmonic mean of precision and recall.
- **AUC-ROC:** Measured the trade-off between sensitivity and specificity.
- **Confusion Matrix:** Provided insights into classification errors, highlighting false positives and negatives.

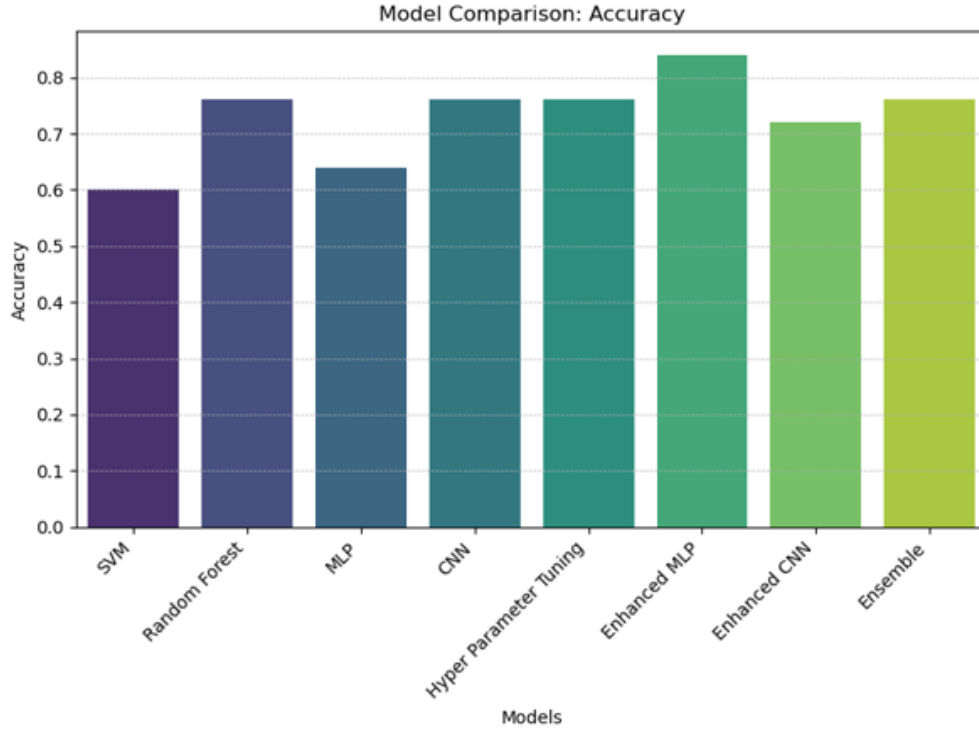


Figure 6: Accuracies for the implementation 1

6.2 Implementation 2: Spectrogram-Based Deep Learning Models

6.2.1 Dataset Description

For this methodology, a dataset comprising 100 voice samples collected via telephone lines was utilized. The samples primarily consisted of sustained vowel sounds, providing a focused perspective on phonation features critical for detecting PD-related impairments Iyer et al. [2023].

6.2.2 Spectrogram Generation

Spectrograms represent the energy distribution of speech signals over time and frequency. The following steps were undertaken:

- **Fast Fourier Transform (FFT):** Used to convert audio signals into frequency domains.
- **Sliding Window:** A window size of 32ms with a 50% overlap ensured high temporal resolution.
- **Visualization:** Both grayscale and colored spectrograms were generated to serve as inputs for deep learning models.

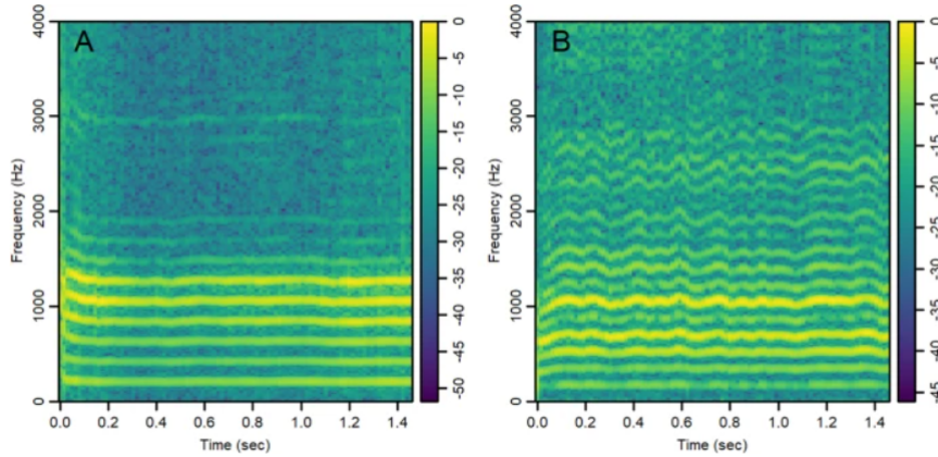


Figure 7: Sample spectral analysis of Healthy and Parkinson's patient

6.2.3 Preprocessing

To ensure the quality of spectrograms, the following preprocessing steps were applied:

- **Noise Reduction:** Background noise was removed using decibel-based filtering.
- **Silence Removal:** Non-informative silent portions were trimmed to standardize the audio durations.
- **Normalization:** The pixel values of spectrograms were normalized to enhance uniformity and comparability across samples.

6.2.4 Model Architecture

A Convolutional Neural Network (CNN) was employed for classification tasks:

- **Inception V3:** A pretrained CNN model, fine-tuned for the task of PD classification.
- **Transfer Learning:** Pretrained weights from ImageNet were leveraged, reducing training time and improving performance.
- **Additional Layers:** Batch normalization and dropout layers were added to mitigate overfitting and enhance generalization.

6.2.5 Model Performance

The CNN achieved an accuracy of **80%**, demonstrating its effectiveness in identifying patterns in spectrograms linked to PD. While spectrogram-based analysis required extensive computational resources, it provided complementary insights to feature-based models.

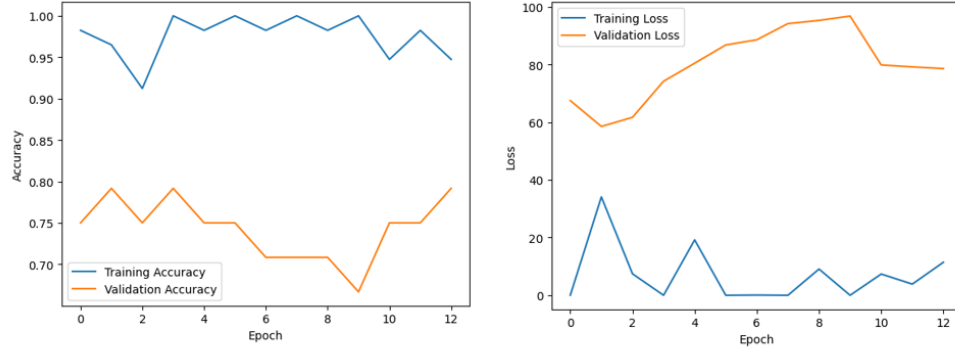


Figure 8: Accuracy and loss for the implementation 2

6.3 Implementation 3: Hybrid Learning Framework

6.3.1 Pipeline Workflow

1. Dataset Usage:

- **Primary Dataset:** 495 recordings from the Italian Parkinson's Voice and Speech Database, encompassing sustained vowel phonations and complex speech tasks.
- **Validation Datasets:**
 - Dataset 1: 34 reading speech samples from English data at King's College London.
 - Dataset 2: 74 vowel samples ("Aah" sounds) from an additional English dataset.

2. Feature Extraction:

Diverse speech features were extracted using Parselmouth (Praat), Librosa, and OpenSMILE. Key features include:

- Parselmouth: Jitter, shimmer, HNR, and formant characteristics.
- Librosa: MFCCs, spectral centroid, ZCR, roll-off, and global statistics (mean, std, min, max).
- OpenSMILE: Features extracted using ComParE 2016 and eGeMAPS configurations, capturing spectral and prosodic attributes.

3. Preprocessing:

- **Feature Reduction:** A correlation matrix was used to remove redundant features with a correlation coefficient $r > 0.9$. Only features with high correlation to the target label were retained, reducing the feature set from 7,208 to 4,026.
- **Standardization:** StandardScaler was applied for consistent feature scaling.

4. **SMOTE-Based Resampling:** To address class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was employed, ensuring equitable class representation in the training set.
5. **Model Training and Evaluation:**
 - **Random Forest:** A versatile ensemble learning method trained with `random_state=42`. Achieved 98% accuracy. Top 20 features were identified using feature importance.
 - **K-Nearest Neighbors (KNN):** Trained with `n_neighbors=5` on standardized data. Permutation importance was computed to rank features, and the top 20 features were extracted. Achieved 94% accuracy.
 - **XGBoost:** A gradient-boosting algorithm trained with hyperparameters like `use_label_encoder=False` and evaluation metric `logloss`. Achieved 99% accuracy.
 - **Feature Analysis:** The top features from all models were aggregated, and their union (52 features) was determined.
6. **Common Feature Selection:** Features overlapping across different models were validated and used for further training.
7. **Final Model Training with LightGBM:** A LightGBM model was trained on the refined feature set of common features. Its performance was evaluated in terms of accuracy and classification metrics.

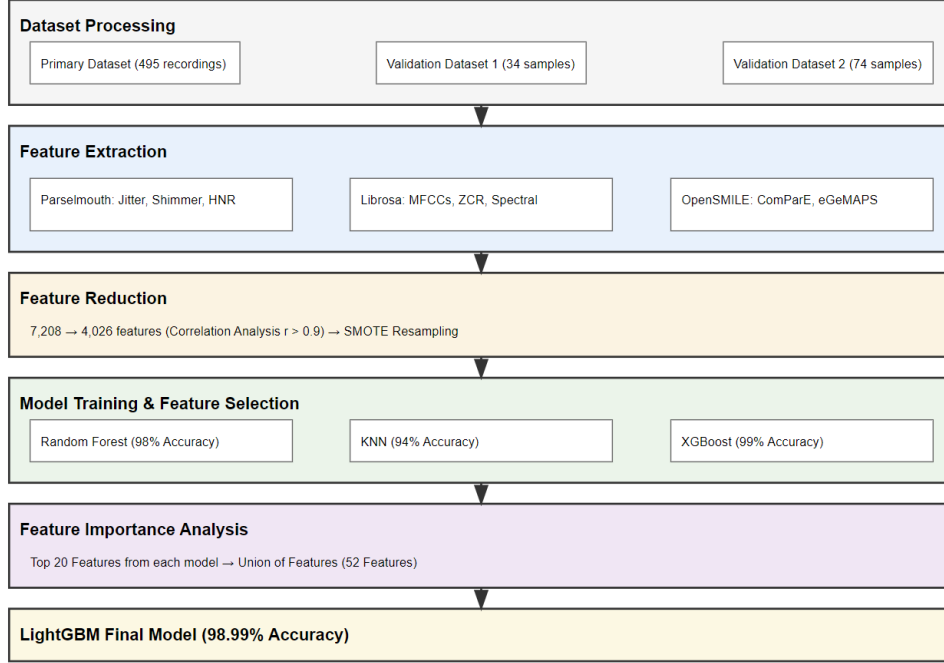


Figure 9: Pipeline workflow diagram showing the complete process from dataset processing through feature extraction, reduction, model training with feature selection, to final LightGBM model deployment. The workflow illustrates the processing of 495 primary recordings and two validation datasets (34 and 74 samples), feature extraction using Parselmouth, Librosa, and OpenSMILE, reduction from 7,208 to 4,026 features, model training achieving 94-99% accuracy across different models, selection of top 20 features from each model leading to 52 unified features, and final LightGBM model achieving 98.99% accuracy.

6.4 Implementation 4: Real-Time Classification Framework

6.4.1 Dataset and Testing Conditions

The real-time framework was initially validated on datasets used in Methodologies 1 and 3, achieving an accuracy of 98%. Further testing on the KCL dataset, which included read text and spontaneous speech tasks, revealed an accuracy of 75%, reflecting the framework's adaptability to diverse data. Here we took all the features from feature set eGeMAPS from opensmile which are 88 features.

6.4.2 Framework Design

The real-time classification framework integrated 16 predictive models:

- **PCA Models:** 8 models trained on features reduced via Principal Component Analysis (PCA) with 95 percent threshold
- **SelectKBest Models:** 8 models trained on features selected using SelectKBest with top 20 features.

6.4.3 Dual Pipelines

- **Pipeline 1:** Feature extraction $\rightarrow$ PCA $\rightarrow$ Model prediction.
- **Pipeline 2:** Feature extraction $\rightarrow$ SelectKBest $\rightarrow$ Model prediction.

6.4.4 Model Performance

The framework's performance demonstrated:

- **High Initial Accuracy:** 98% accuracy on initial datasets.

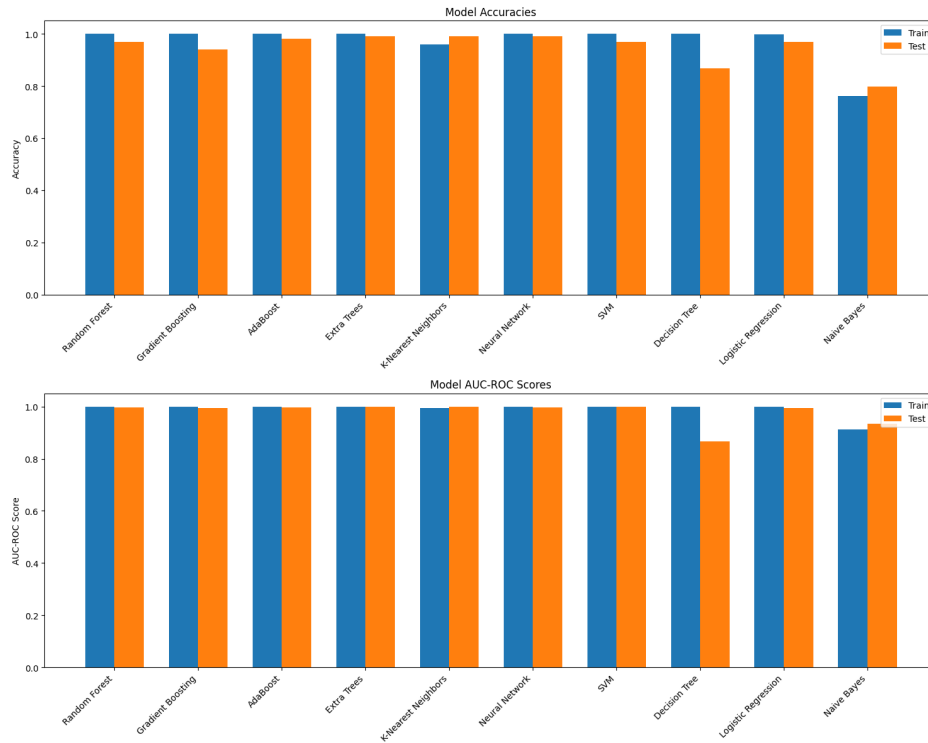


Figure 10: Accuracy through the PCA pipeline

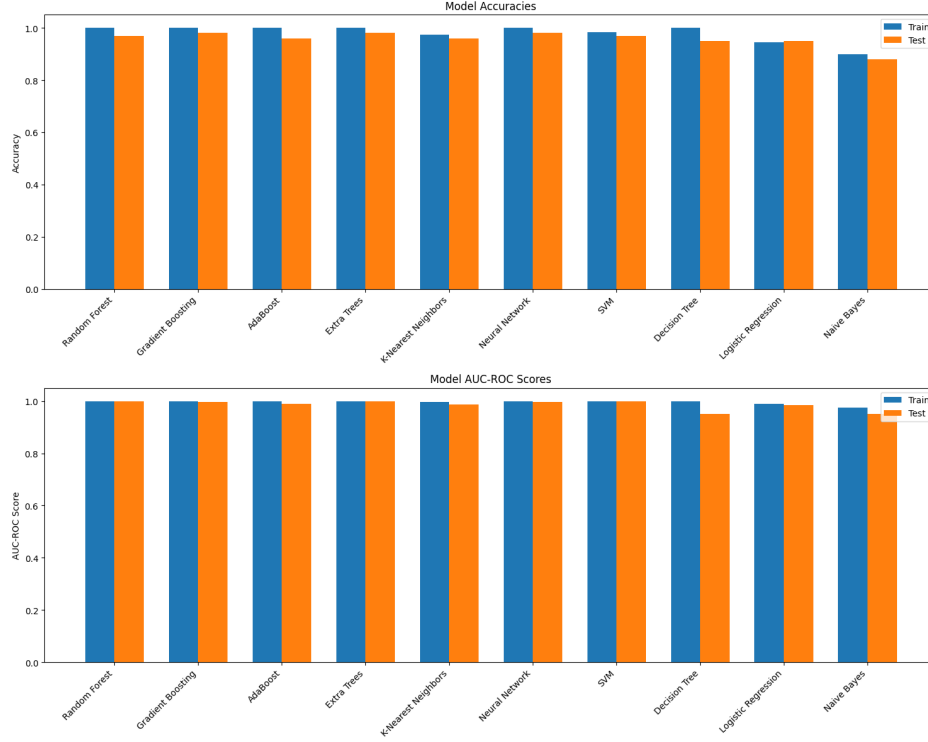


Figure 11: Accuracy through the Select K best pipeline

- **Generalizability:** 75% accuracy on diverse and unseen data from the KCL dataset.

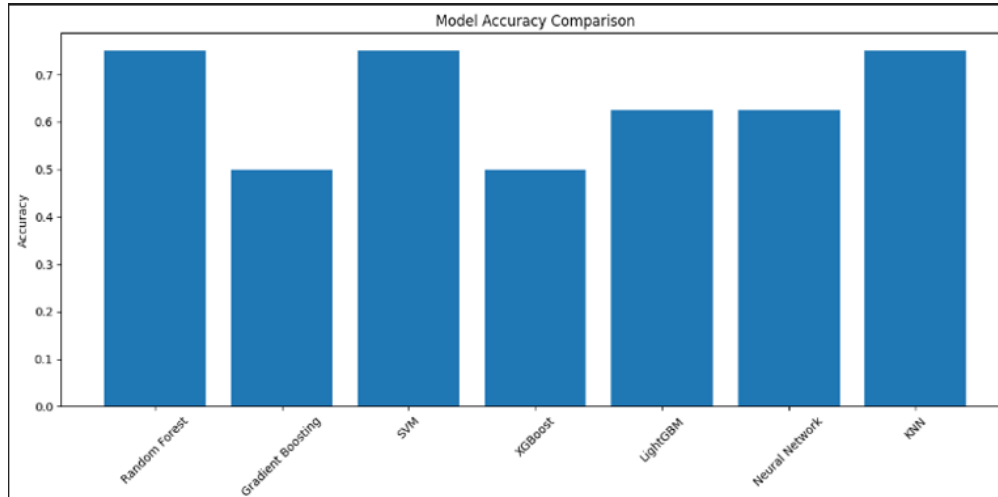


Figure 12: Model accuracy for readtext data

6.4.5 Real-Time Functionality

The framework processed live audio clips through the following steps:

1. **Feature Extraction:** Acoustic features were extracted using OpenSMILE in real time.

2. **Model Predictions:** Each of the 16 models provided predictions and confidence scores.
3. **Consensus Mechanism:** Aggregated predictions into a final decision, ensuring robustness and reliability.
4. **Feature Importance:** Highlighted the top five features influencing predictions, improving interpretability.

Parkinson's Disease Voice Classification

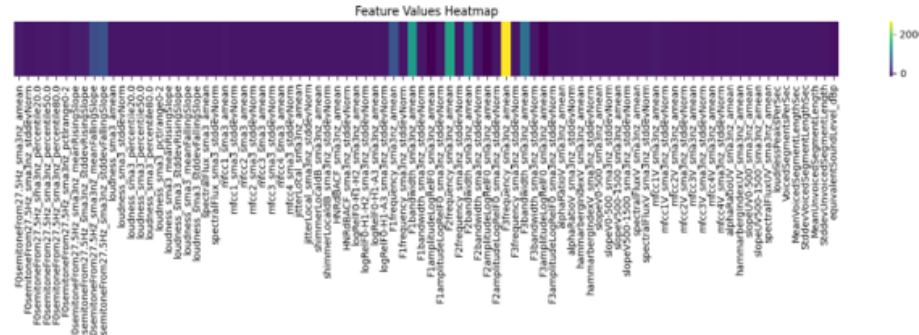
Click to record your voice for 5 seconds.

Record (5 sec)

Your Recordings

0:00 / 0:05

Extracted Features:



Classification Results:

Overall Consensus:

Healthy (Agreement: 62.5%)

Average Confidence: 92.3%

SelectKBest - KNN

Prediction: Healthy

Confidence: 100.00%

SelectKBest - Decision Tree

Prediction: Healthy

Confidence: 100.00%

PCA - KNN

Prediction: Healthy

Confidence: 100.00%

PCA - Decision Tree

Prediction: Healthy

Confidence: 100.00%

Figure 13: Real time dashboard

7 Code Implementation

Key parts of the implementation leverage specialized libraries for feature extraction and processing:

```
from opensmile import Smile, FeatureSet, FeatureLevel
from parselmouth import Sound
import librosa

# OpenSMILE Feature Extraction
smile = Smile(feature_set=FeatureSet.ComParE_2016, feature_level=FeatureLevel.Functionals)
smile_features = smile.process_file('audio_file.wav')

# Librosa Feature Extraction
mfccs = librosa.feature.mfcc(y=audio_data, sr=sample_rate, n_mfcc=13)

# Parselmouth Feature Extraction
sound = Sound('audio_file.wav')
pitch = sound.to_pitch()
mean_pitch = pitch.selected_array['frequency'].mean()
```

8 Results and Discussion

8.1 Comparative Analysis

The following table compares the performance of different models, highlighting results from reduced features (our optimized models) and their respective datasets:

Model (Dataset)	Accuracy (%)	AUC
LightGBM	98.99	0.99
Random Forest	97.97	0.98
Adaboost	98.94	0.94
XGBoost	98.98	0.99
MLP	98.31	0.98
SVM	95.00	0.95
Random Forest	93.00	0.94
KNN	93.20	0.88

Table 1: Performance Metrics of Different Models selected Features)

8.2 Validation on External Datasets

The model performance was validated on external datasets, as described below:

- **Dataset (English Dataset, "Aah" Vowel Sounds):**
 - **Samples:** 74 total (37 Healthy, 37 Parkinson's).
 - **Accuracy:** 76.0%.
 - **Remarks:** Despite linguistic and phonetic differences, the model showed robust accuracy.

8.3 Discussion

8.3.1 Model Performance and Comparison

- **LightGBM and XGBoost:** These models performed exceptionally well, leveraging the refined feature set and optimized hyperparameters. The high AUC values (0.99) reflect excellent separation between HC and PD classes.
- **MLP:** The Multi-Layer Perceptron demonstrated competitive performance (98.31% accuracy), highlighting its ability to capture complex feature interactions. However, its training time was higher compared to tree-based methods.
- **Spectrogram-Based CNN:** Achieved an accuracy of 80% using Inception V3, showcasing the potential of image-based deep learning for PD classification. However, this method requires significant computational resources and preprocessing time.
- **Real-Time Framework:** Demonstrated robust performance, achieving 98% accuracy in initial testing and 75% on unseen datasets. The dual pipeline approach ensures adaptability, but the reduced accuracy on KCL data suggests room for improvement in handling diverse recording conditions.

8.3.2 Validation and Generalizability

- The relatively lower accuracy on Dataset 2 ("Aah" vowel sounds) can be attributed to linguistic and acoustic variations, suggesting potential areas for further feature refinement. Despite this, the model maintained strong predictive performance, confirming its robustness.

8.3.3 Feature Importance Analysis

Key features contributing to classification included:

- **MDVP:F0 (Hz):** Indicative of frequency variability.
- **jitterLocal_sma3nz_amean:** Mean local jitter in normalized speech signal.
- **shimmerLocal_sma3nz_stddevNorm:** Standard deviation of local shimmer (amplitude irregularities).
- **HNRdBACF_sma3nz_amean:** Mean Harmonics-to-Noise Ratio (HNR) in decibels.
- **pcm_fftMag_fband250-650_sma3_amean:** Mean spectral magnitude in the frequency band 250-650 Hz.
- **voicingFinalUnclipped_sma3nz_mean:** Mean voicing probability (non-clipped signal).
- **F0semitoneFrom27.5Hz_sma3nz_amean:** Mean semitone from 27.5 Hz fundamental frequency.
- **MFCC:** Critical spectral features for speech pattern differentiation.

8.3.4 Comparison with Existing Studies

This research demonstrates significant improvements over traditional methods by incorporating advanced preprocessing and feature selection techniques. Existing literature often focuses on limited datasets and single-feature analyses, which restrict generalizability. In contrast, this framework leverages diverse feature sets, enhancing robustness and applicability.

8.4 Challenges and Limitations

1. **Data Imbalance:** Despite SMOTE resampling, the limited diversity of the datasets may introduce biases, especially for underrepresented linguistic or demographic groups.
2. **Noise Sensitivity:** Background noise and varying recording environments can affect feature extraction quality. Future work should focus on robust preprocessing techniques to mitigate these issues.
3. **Linguistic and Acoustic Variability:** Models trained on specific datasets struggled to generalize across languages and recording contexts. Incorporating diverse datasets will improve cross-linguistic robustness.
4. **Computational Demand:** Spectrogram-based methods, while effective, require high computational power and are less practical for real-time applications compared to feature-based models.
5. **Feature Reduction Trade-Off:** While dimensionality reduction improved efficiency, there's a risk of discarding potentially valuable features. Iterative feature selection and validation can mitigate this risk.
6. **Dataset Size and Diversity Limitations:** The primary limitation of this study lies in the relatively small size of available datasets, particularly for cross-linguistic validation. While the initial results are promising, the limited sample size may not fully capture the vast spectrum of voice variations in Parkinson's Disease patients across different demographics and disease stages. The model's performance might be constrained by the lack of diverse training examples, potentially affecting its generalizability to broader populations.
7. **External Dataset Validation Challenges:** The observed decrease in accuracy when testing on external datasets, particularly the English "AHH"sounds dataset, highlights challenges in cross-dataset generalization. The authenticity and recording conditions of external datasets cannot be fully verified, introducing potential confounding variables. Variations in data collection protocols, recording equipment, and environmental conditions between datasets may impact model performance. This is evidenced by the drop in accuracy from 98.99% on our primary dataset to 75% on external validation datasets.
8. **Model Generalization Issues:** The performance disparity between internal validation and external testing suggests that the model may be capturing dataset-specific patterns rather than universally applicable features. The current feature set, while effective for the primary dataset, may not fully capture the acoustic characteristics that generalize across different languages and recording conditions.

9 Conclusion

This study underscores the transformative potential of speech analysis in Parkinson’s Disease diagnostics. By combining feature-based ML, spectrogram-based CNNs, and real-time frameworks, this research achieves high accuracy and adaptability, paving the way for scalable clinical solutions. Further validation on diverse datasets and integration with other modalities will enhance the robustness and clinical relevance of the proposed methodologies. Tsanas et al. [2012].

Key Takeaways

1. **Optimized ML Models:** LightGBM and XGBoost outperformed other methods in terms of accuracy and computational efficiency.
2. **Deep Learning Potential:** Spectrogram-based CNNs offer promising avenues for PD detection but require further optimization for real-time use.
3. **Real-Time Feasibility:** The proposed dual-pipeline framework demonstrated the potential for live applications, with aggregated predictions ensuring robustness.

10 Future Work

1. **Multimodal Biomarker Integration:** Combining speech analysis with other PD biomarkers such as gait and handwriting patterns The Parkinson’s Progression Markers Initiative (PPMI).
2. **Real-Time Monitoring Devices:** Developing mobile or wearable solutions for continuous monitoring.
3. **Cross-Language Generalizability:** Expanding datasets to include multiple languages.
4. **Explainable AI:** Enhancing model interpretability for clinical adoption.
5. **Dataset Augmentation:** Leveraging synthetic data generation to overcome dataset limitations Sriram et al. [2014].

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